COMP3516 Group Project

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ABSTRACT

<Write the abstract>

1 INTRODUCTION

<Write the Introduction>

2 TECHNICAL PART

2.1 Hardware setups

2.2 Breathing rate estimation algorithm

2.3 Motion detection algorithm

The motion detection via CSI data is based on the statistical electromagnetic model proposed by [1]. For conciseness, we omit the details of the electromagnetic model, but focus on the motion detection based on the motion statistics.

Suppose the CSI data collected is indexed by $H(t, f)$, while we omit other parameters in the dataframe. The corresponding power response $G(t, f) = |H(f, t)|^2$.

We use the sample auto-covariance function $\hat{\gamma}_G(\tau, f)$ where τ is the lag (c.f. Eqn. (2) in [1]), which is defined as

$$\hat{\gamma}_G(\tau, f) = \frac{1}{T} \sum_{t=\tau+1}^{T} (G(t-\tau, f) - \overline{G}(f))(G(t, f) - \overline{G}(f)),$$

where T is the number of samples (within a window). The ACF of $G(t, f)$ characterized by the lag $\tau > 0$ is defined as

$$\rho_G(\tau, f) = \frac{\hat{\gamma}_G(\tau, f)}{\hat{\gamma}_G(0, f)}.$$

Whenever the calculation of $\rho_G(\tau, f)$ fails, we discard this subcarrier. The motion detection is performed by hypothesis testing. Denote the set of subcarriers as $\mathcal{F}$, for each $f \in \mathcal{F}$, define the **motion statistic** as

$$\phi(f) = \rho_G\left(\frac{1}{F_s}, f\right).$$

To perform the following hypothesis testing:

$$\begin{aligned} H_0 : &\quad \hat{\phi}(f) \sim \mathcal{N}(-1/T, 1/T), \quad \forall f \in \mathcal{F}, \\ H_1 : &\quad \hat{\phi}(f) \xrightarrow{\tau \to 0} \eta(f) > 0, \quad \forall f \in \mathcal{F}. \end{aligned}$$

We use the motion statistic estimator

$$\hat{\psi} = \frac{1}{|\mathcal{F}|} \sum_{f \in \mathcal{F}} \hat{\phi}(f)$$

to detect the motion, and the decision rule for motion (1) or static (0) is given by $\mathbb{I}\left\{\hat{\psi} > \eta\right\}$ for some appropriate choice of parameter η.

3 EVALUATION

3.1 Evaluation Result

<Write your analysis>

Table 1: Overall Evaluation Result.

Evaluation Dataset	Result
Motion Detection	awaiting test
Breathing Rate Estimation	<Your median MAE (BPM)>

3.2 Benchmark Result

3.2.1 *Breathing Rate Test.* <Write your analysis>

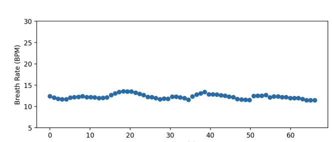

Figure 1: Estimated Respiration for 193342.csv.

*Equal contribution.

Table 2: Test result of motion detection with ground truth motion.

File Name	Result	File Name	Result
CSI_20250220_204932.csv	100%	CSI_20250220_205516.csv	100%
CSI_20250220_205301.csv	100%	CSI_20250220_205526.csv	100%
CSI_20250220_205318.csv	100%	CSI_20250220_205539.csv	100%
CSI_20250220_205330.csv	100%	CSI_20250220_205553.csv	100%
CSI_20250220_205343.csv	100%	CSI_20250220_205607.csv	100%
CSI_20250220_205356.csv	100%	CSI_20250220_205624.csv	100%
CSI_20250220_205406.csv	100%	CSI_20250220_205637.csv	100%
CSI_20250220_205439.csv	80%	CSI_20250220_205645.csv	100%
CSI_20250220_205451.csv	100%	CSI_20250220_205654.csv	100%
CSI_20250220_205504.csv	100%	CSI_20250220_205704.csv	100%

Table 3: Test result of motion detection with ground truth static.

File Name	Result	File Name	Result
CSI_20250220_203408.csv	100%	CSI_20250220_210747.csv	100%
CSI_20250220_204824.csv	100%	CSI_20250220_210802.csv	100%
CSI_20250220_210608.csv	100%	CSI_20250220_210814.csv	100%
CSI_20250220_210623.csv	80%	CSI_20250220_210840.csv	100%
CSI_20250220_210645.csv	100%	CSI_20250220_210854.csv	100%

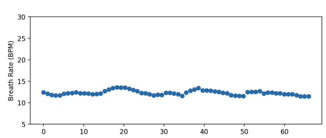

Figure 2: Estimated Respiration for 200223.csv.

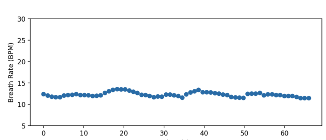

Figure 3: Estimated Respiration for 201424.csv.

3.2.2 Motion Test. We conduct the motion detection test on all the csv files contained in directory `benchmark/motion_detection/evaluation_motion` and `evaluation_static`. As each csv contains 300 samples, the sampling rate F_s is 100Hz, we set the windows size to be 100, step size to be 50, the lag $\tau = 1$, and the threshold $\eta = 0.4$. The algorithm reports an accuracy of 98% on static benchmark and 99% on motion benchmark. Accuracy on each testcase is reported in Table 2 and Table 3.

4 CONCLUSION

A APPENDIX I

A.1 Individual Contribution

A.2 Checklist of your submission

B APPENDIX II

REFERENCES

[1] Feng Zhang, Chenshu Wu, Beibei Wang, Hung-Quoc Lai, Yi Han, and K. J. Ray Liu. 2019. WiDetect: Robust Motion Detection with a Statistical Electromagnetic Model. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 3, 3, Article 122 (Sept. 2019), 24 pages. https://doi.org/10.1145/3351280

Table 4: Individual Contribution

Name	UID	Contribution Statement
A	a	a%, a's role and contributions here.
B	b	b%, b's role and contributions here.
C	c	c%, c's role and contributions here.
D	d	d%, d's role and contributions here.

Table 5: Self-Report Submission Checklist (1=Done, 0=No)

Items	Status
Successfully receive the CSI data.	__
Successfully transmit the data over MQTT.	__
The algorithm is implemented on-board.	__
The algorithm is implemented in real-time.	__
Successfully develop an end-to-end system for data visualization.	__
Submit the video (no more than 3 minutes).	__
The submitted files contain all code necessary to reproduce and verify the results, ensuring full reproducibility of the work.	__
We confirm that all data results are consistent with the code execution outcomes.	__
We confirm that the contribution statement is accurate and that there are no objections from any team members.	__
We acknowledge the use of GenAI and ensure that it has been correctly cited or referenced in accordance with the required guidelines.	__
We certify that all information in this checklist is accurate, complete, and compliant with academic integrity policies.	__